

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
knsat@SRITHIKA: ~
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```
knsat@SRITHIKA: ~  
knsat@SRITHIKA:~$ nano pipe_cmd.c  
knsat@SRITHIKA:~$ gcc pipe_cmd.c -o pipe_cmd  
knsat@SRITHIKA:~$ ./pipe_cmd  
drwxr-xr-x 2 knsat knsat 4096 Sep 19 04:56 IPC_Process_Management  
-rw-r--r-- 1 knsat knsat 576 Aug 27 03:32 ano.c  
-rwxr-xr-x 1 knsat knsat 16168 Aug 24 08:10 copy  
-rw-r--r-- 1 knsat knsat 515 Aug 24 08:32 copy.c  
-rwxr-xr-x 1 knsat knsat 16048 Sep 2 05:39 create_fifo  
-rw-r--r-- 1 knsat knsat 221 Sep 2 05:46 create_fifo.c  
-rw-r--r-- 1 knsat knsat 111 Aug 12 03:53 display.c  
-rwxr-xr-x 1 knsat knsat 16272 Aug 24 04:08 exec_command  
-rw-r--r-- 1 knsat knsat 439 Aug 24 03:54 exec_command.c  
-rwxr-xr-x 1 knsat knsat 16049 Aug 27 04:14 fifo_create  
-rw-r--r-- 1 knsat knsat 218 Aug 27 04:00 fifo_create.c  
-rw-r--r-- 1 knsat knsat 253 Aug 27 04:14 fifo_createread.c  
-rwxr-xr-x 1 knsat knsat 16184 Aug 27 04:20 fifo_createwrite  
-rw-r--r-- 1 knsat knsat 292 Aug 27 04:20 fifo_createwrite.c  
-rw-r--r-- 1 knsat knsat 253 Sep 2 05:39 fifo_read.c  
-rw-r--r-- 1 knsat knsat 255 Sep 2 05:57 fifo_read.c.save  
-rw-r--r-- 1 knsat knsat 292 Sep 2 05:39 fifo_write.c  
-rw-r--r-- 1 knsat knsat 255 Sep 26 05:59 fiforead.c  
-rw-r--r-- 1 knsat knsat 292 Sep 26 06:35 fifowrite.c  
-rw----- 1 knsat knsat 159 Aug 6 03:56 file.c.save  
-rw-r--r-- 1 knsat knsat 91 Aug 13 04:03 fork.c  
-rwxr-xr-x 1 knsat knsat 16000 Aug 12 04:07 forkcalling  
-rw-r--r-- 1 knsat knsat 142 Aug 12 04:07 forkcalling.c  
drwxr-xr-x 3 knsat knsat 4096 Sep 19 04:58 github_sync  
-rw----- 1 knsat knsat 436 Aug 6 05:18 hallo.c.save  
-rw-r--r-- 1 knsat knsat 174 Aug 1 05:53 hello.c  
-rw-r--r-- 1 knsat knsat 239 Aug 22 04:54 orphan.c  
-rw-r--r-- 1 knsat knsat 296 Aug 26 05:34 p2.c  
-rw-r--r-- 1 knsat knsat 335 Aug 26 05:36 p3.c  
-rw-r--r-- 1 knsat knsat 230 Aug 26 05:38 p4.c  
-rw-r--r-- 1 knsat knsat 179 Aug 26 05:43 p5.c  
-rw-r--r-- 1 knsat knsat 194 Aug 26 05:44 p6.c  
-rw-r--r-- 1 knsat knsat 314 Aug 26 05:51 p7.c  
-rwxr-xr-x 1 knsat knsat 16216 Sep 26 06:37 pipe_cmd  
-rw-r--r-- 1 knsat knsat 498 Sep 26 06:36 pipe_cmd.c  
-rw-r--r-- 1 knsat knsat 122 Aug 26 05:32 pl.c  
drwxr-xr-x 2 knsat knsat 4096 Jul 30 03:42 practical  
-rwxr-xr-x 1 knsat knsat 16000 Aug 25 03:59 process  
-rw-r--r-- 1 knsat knsat 112 Aug 25 03:59 process.c  
drwxr-xr-x 2 knsat knsat 4096 Jul 30 03:42 projects  
-rw-r--r-- 1 knsat knsat 254 Aug 27 04:14 read.c  
-rw-r--r-- 1 knsat knsat 137 Aug 12 03:41 readusername.c  
-rwxr-xr-x 1 knsat knsat 16192 Aug 25 04:50 simplecommandloop  
-rw-r--r-- 1 knsat knsat 609 Aug 25 04:49 simplecommandloop.c  
-rw-r--r-- 1 knsat knsat 1037 Aug 26 08:28 skill3.c
```

2.Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

GNU nano 8.7.1

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```
knsat@SRITHIKA:~$ nano signal_demo.c
knsat@SRITHIKA:~$ gcc signal_demo.c -o signal_demo
knsat@SRITHIKA:~$ ./signal_demo
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
^C
SIGINT Received
Running...
Running...
Running...
Running...
Running...
Running...
```

```
knsat@SRITHIKA: ~
knsat@SRITHIKA:~$ pidof signal_demo
1416
knsat@SRITHIKA:~$ kill -SIGUSR1 12345
-bash: kill: (12345) - No such process
knsat@SRITHIKA:~$ kill -SIGUSR1 1416
knsat@SRITHIKA:~$ kill -SIGUSR1 1416
knsat@SRITHIKA:~$ kill -9 1416
```